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GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Numerical Methods

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

11	11	1993 – 2017	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	6 questions
Marks 2	5 questions
Answer key	at the end

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Marks 1 6 questions

Q1. GATE ECE 2015 Set 3

The Newton-Raphson method is used to solve the equation $f(x) = x^3 - 5x^2 + 6x - 8 = 0$. Taking the initial guess as $x = 5$, the solution obtained at the end of the first iteration is _____.

Open solution

Q2. GATE ECE 2011

A numerical solution of the equation $f(x) = x + \sqrt{x} - 3 = 0$ can be obtained using Newton $-$ Raphson method. If the starting values is $x = 2$ for the iteration then the value of x that is to be used in the next step is

(A) 0.306 (B) 0.739 (C) 1.694 (D) 2.306

Open solution

Q3. GATE ECE 2010

Consider a differential equation $\frac{dy(x)}{dx} - y(x) = x$ with initial condition $y(0) = 0$. Using Euler's first order method with a step size of 0.1 then the value of $y(0.3)$ is

(A) 0.01 (B) 0.031 (C) 0.0631 (D) 0.1

Open solution

Q4. GATE ECE 2008

The recursion relation to solve $x = e^{-x}$ using Newton $-$ Raphson method is

(A) $x_{n+1} = e^{-x_n}$

(B) $x_{n+1} = x_n - e^{-x_n}$

(C) $x_{n+1} = \frac{(1+x_n)e^{-x_n}}{(1+e^{-x_n})}$

(D) $x_{n+1} = \frac{x_n^2 - e^{-x_n}(1+x_n) - 1}{x_n - e^{-x_n}}$

Open solution

Q5. GATE ECE 2007

The equation $x^3 - x^2 + 4x - 4 = 0$ is to be solved using the Newton - Raphson method. If $x = 2$ taken as the initial approximation of the solution then the next approximation using this method, will be

(A) $2/3$ (B) $4/3$ (C) 1 (D) $3/2$

Open solution

Q6. GATE ECE 1993

Given the differential equation $y^1 = x - y$ with initial condition $y(0) = 0$. The value of $y(0.1)$ calculated numerically upto the third place of decimal by the 2nd order Runge-Kutta method with step size $h = 0.1$ is

Open solution

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Marks 2 5 questions

Q7. GATE ECE 2017 Set 1

Starting with $x = 1$, the solution of the equation $x^3 + x = 1$, after two iterations of Newton-Raphson's method (up to two decimal places) is _____

Open solution

Q8. GATE ECE 2016 Set 2

The ordinary differential equation $\frac{dx}{dt} = -3x + 2$, with $x(0) = 1$ is to be solved using the forward Euler method. The largest time step that can be used to solve the equation without making the numerical solution unstable is _____.

Open solution

Q9. GATE ECE 2016 Set 3

Consider the first order initial value problem $y' = y + 2x - x^2$, $y(0) = 1$, $(0 \leq x < \infty)$ With exact solution $y(x) = x^2 + e^x$. For $x = 0.1$, the percentage difference between the exact solution and the solution obtained using a single iteration of the second-order Runge-Kutta method with step-size $h = 0.1$ is _____.

Open solution

Q10. GATE ECE 2014 Set 3

Match the application to appropriate numerical method

Applications

$P1$: Numerical integration
$P2$: Solution to a transcendental equation
$P3$: Solution to a system of linear equations
$P4$: Solution to a differential equation

Numerical Method

$M1$: Newton-Raphson Method
$M2$: Runge-Kutta Method
$M3$: Simpson's $1/3-$rule
$M4$: Gauss Elimination Method

(A) $P1 - M3,\ P2 - M2,\ P3 - M4,\ P4 - M1$
(B) $P1 - M3,\ P2 - M1,\ P3 - M4,\ P4 - M2$
(C) $P1 - M4,\ P2 - M1,\ P3 - M3,\ P4 - M2$
(D) $P1 - M2,\ P2 - M1,\ P3 - M3,\ P4 - M4$

Open solution

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Q11. GATE ECE 2005

Open solution

Match the following and choose the correct combination

Group $-$ I

E. Newton $-$ Raphson method

F. Runge-Kutta method

G. Simpson's Rule

H. Gauss elimination

Group $-$ II

(1) Solving non-linear equations

(2) Solving linear simultaneous equations

(3) Solving ordinary differential equations

(4) Numerical integration method

(5) Interpolation

(6) Calculation of eigen values

(A) $E-6,\ F-1,\ G-5,\ H-3$

(B) $E-1,\ F-6,\ G-4,\ H-3$

(C) $E-1,\ F-3,\ G-4,\ H-2$

(D) $E-5,\ F-3,\ G-4,\ H-1$

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Answer Key

Marks 1

1. **4.3**	2. **C**	3. **B**	4. **C**	5. **B**
6. $\frac{1}{200}$				

Marks 2

7. **0.69**	8. **0.66 to 0.67**	9. **0.06**	10. **B**	11. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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